```sql
--- CREATE DATABASE
===================================
CREATE DATABASE Techno1;
USE Techno1;

/*=======================================================
  CREATE DEPARTMENT TABLE
=======================================================*/

CREATE TABLE Department
(
   DeptID INT PRIMARY KEY,           -- Unique Department ID
   DeptName VARCHAR(50) NOT NULL     -- Department Name
);


/*=======================================================
   CREATE EMPLOYEE TABLE
=======================================================*/

CREATE TABLE Employee
(
   EmpID INT PRIMARY KEY,               -- Unique Employee ID
   EmpName VARCHAR(50) NOT NULL,        -- Employee Name
   Salary DECIMAL(10,2),                -- Employee Salary
   DeptID INT,                          -- Department ID (Foreign Key)
   HireDate DATE,                       -- Employee Joining Date

   FOREIGN KEY (DeptID)
   REFERENCES Department(DeptID)
);


/*=======================================================
   INSERT DATA INTO DEPARTMENT TABLE
=======================================================*/

INSERT INTO Department (DeptID, DeptName)
VALUES
(1, 'HR'),
(2, 'IT'),
(3, 'Finance'),
(4, 'Sales'),
(5, 'Marketing'),
```

```
(6, 'Admin'),
(7, 'Operations'),
(8, 'Support'),
(9, 'Legal'),
(10, 'Research');


/*=======================================================
   INSERT DATA INTO EMPLOYEE TABLE
=======================================================*/

INSERT INTO Employee (EmpID, EmpName, Salary, DeptID, HireDate)
VALUES
(101, 'Megha', 75000, 2, '2026-02-10'),
(102, 'Rahul', 65000, 1, '2025-12-15'),
(103, 'Amit', 90000, 2, '2026-01-20'),
(104, 'Priya', 85000, 3, '2025-11-10'),
(105, 'Sneha', 60000, 4, '2026-03-05'),
(106, 'Vikas', 95000, 2, '2025-10-01'),
(107, 'Neha', 70000, 5, '2026-04-12'),
(108, 'Kiran', 55000, 1, '2026-05-01'),
(109, 'Rohan', 80000, 3, '2026-01-15'),
(110, 'Pooja', 68000, 4, '2026-06-01');


/*=======================================================
   DISPLAY ALL RECORDS
=======================================================*/

-- View Department Table
SELECT *
FROM Department;

-- View Employee Table
SELECT *
FROM Employee;
```

Results | Messages

	DeptID	DeptName
1	1	HR
2	2	IT
3	3	Finance
4	4	Sales
5	5	Marketing
6	6	Admin
7	7	Operations
8	8	Support
9	9	Legal
10	10	Research

	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	102	Rahul	65000.00	1	2025-12-15
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	105	Sneha	60000.00	4	2026-03-05
6	106	Vikas	95000.00	2	2025-10-01
7	107	Neha	70000.00	5	2026-04-12
8	108	Kiran	55000.00	1	2026-05-01
9	109	Rohan	80000.00	3	2026-01-15
10	110	Pooja	68000.00	4	2026-06-01

```
/*=====================================================
   1. TOP 5 HIGHEST SALARY EMPLOYEES
=====================================================*/

SELECT TOP 5 *
FROM Employee
ORDER BY Salary DESC;
```

	EmpID	EmpName	Salary	DeptID	HireDate
1	106	Vikas	95000.00	2	2025-10-01
2	103	Amit	90000.00	2	2026-01-20
3	104	Priya	85000.00	3	2025-11-10
4	109	Rohan	80000.00	3	2026-01-15
5	101	Megha	75000.00	2	2026-02-10

```
/*=======================================================
  2. DEPARTMENT-WISE EMPLOYEE COUNT
=======================================================*/

SELECT
    d.DeptName,
    COUNT(e.EmpID) AS EmployeeCount
FROM Department d
INNER JOIN Employee e
    ON d.DeptID = e.DeptID
GROUP BY d.DeptName;
```

Results Messages

	DeptName	EmployeeCount
1	Finance	2
2	HR	2
3	IT	3
4	Marketing	1
5	Sales	2

```
/*=======================================================
  3. SECOND HIGHEST SALARY
=======================================================*/

SELECT MAX(Salary) AS SecondHighestSalary
FROM Employee
WHERE Salary <
(
    SELECT MAX(Salary)
    FROM Employee
);
```

	SecondHighestSalary
1	90000.00

```
/*=======================================================
  4. EMPLOYEES WHOSE SALARY IS GREATER
     THAN THEIR DEPARTMENT AVERAGE
=======================================================*/

SELECT *
FROM Employee e
```

```
WHERE Salary >
(
    SELECT AVG(Salary)
    FROM Employee
    WHERE DeptID = e.DeptID
);
```

Results Messages

	EmpID	EmpName	Salary	DeptID	HireDate
1	102	Rahul	65000.00	1	2025-12-15
2	106	Vikas	95000.00	2	2025-10-01
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	110	Pooja	68000.00	4	2026-06-01

```
/*=======================================================
    5. INNER JOIN
=======================================================*/

SELECT
    e.EmpID,
    e.EmpName,
    d.DeptName,
    e.Salary
FROM Employee e
INNER JOIN Department d
ON e.DeptID = d.DeptID;
```

Results | Messages

	EmpID	EmpName	DeptName	Salary
1	101	Megha	IT	75000.00
2	102	Rahul	HR	65000.00
3	103	Amit	IT	90000.00
4	104	Priya	Finance	85000.00
5	105	Sneha	Sales	60000.00
6	106	Vikas	IT	95000.00
7	107	Neha	Marketing	70000.00
8	108	Kiran	HR	55000.00
9	109	Rohan	Finance	80000.00
10	110	Pooja	Sales	68000.00

```
/*=====================================================
  6. LEFT JOIN
=====================================================*/

SELECT
    d.DeptID,
    d.DeptName,
    e.EmpName
FROM Department d
LEFT JOIN Employee e
ON d.DeptID = e.DeptID;


/*=====================================================
    7. GROUP BY WITH HAVING
    DEPARTMENTS HAVING MORE THAN 1 EMPLOYEE
=====================================================*/

SELECT
    DeptID,
    COUNT(*) AS TotalEmployees
FROM Employee
GROUP BY DeptID
HAVING COUNT(*) > 1;
```

	DeptID	DeptName	EmpName
1	1	HR	Rahul
2	1	HR	Kiran
3	2	IT	Megha
4	2	IT	Amit
5	2	IT	Vikas
6	3	Finance	Priya
7	3	Finance	Rohan
8	4	Sales	Sneha
9	4	Sales	Pooja
10	5	Marketing	Neha
11	6	Admin	NULL
12	7	Operations	NULL
13	8	Support	NULL
14	9	Legal	NULL
15	10	Research	NULL

```
/*=======================================================
  7. GROUP BY WITH HAVING
  DEPARTMENTS HAVING MORE THAN 1 EMPLOYEE
=======================================================*/

SELECT
    DeptID,
    COUNT(*) AS TotalEmployees
FROM Employee
GROUP BY DeptID
HAVING COUNT(*) > 1;
```

Results | Messages

	DeptID	TotalEmployees
1	1	2
2	2	3
3	3	2
4	4	2

```
/*=======================================================
  8. EMPLOYEES HIRED IN LAST 6 MONTHS
=======================================================*/

SELECT *
```

```sql
FROM Employee
WHERE HireDate >= DATEADD(MONTH, -6, GETDATE());
```

Results | Messages

	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	103	Amit	90000.00	2	2026-01-20
3	105	Sneha	60000.00	4	2026-03-05
4	107	Neha	70000.00	5	2026-04-12
5	108	Kiran	55000.00	1	2026-05-01
6	109	Rohan	80000.00	3	2026-01-15
7	110	Pooja	68000.00	4	2026-06-01

```sql
/*=====================================================
   9. INSERT DUPLICATE RECORDS FOR PRACTICE
=====================================================*/

INSERT INTO Employee
VALUES
(111, 'Megha', 75000, 2, '2025-02-10'),
(112, 'Rahul', 65000, 1, '2024-12-15');
```

Messages

```
(2 rows affected)

Completion time: 2026-06-23T15:02:11.9122031+05:30
```

	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	102	Rahul	65000.00	1	2025-12-15
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	105	Sneha	60000.00	4	2026-03-05
6	106	Vikas	95000.00	2	2025-10-01
7	107	Neha	70000.00	5	2026-04-12
8	108	Kiran	55000.00	1	2026-05-01
9	109	Rohan	80000.00	3	2026-01-15
10	110	Pooja	68000.00	4	2026-06-01
11	111	Megha	75000.00	2	2025-02-10
12	112	Rahul	65000.00	1	2024-12-15

```
/*=======================================================
  10.  FIND DUPLICATE RECORDS
=======================================================*/

SELECT
    EmpName,
    Salary,
    COUNT(*) AS DuplicateCount
FROM Employee
GROUP BY EmpName, Salary
HAVING COUNT(*) > 1;
```

Results Messages

	EmpName	Salary	DuplicateCount
1	Rahul	65000.00	2
2	Megha	75000.00	2

```
/*=======================================================
   11. REMOVE DUPLICATE RECORDS
=======================================================*/

WITH DuplicateRecords AS
(
    SELECT *,
          ROW_NUMBER() OVER
          (
```

```
            PARTITION BY EmpName, Salary
            ORDER BY EmpID
        ) AS RN
    FROM Employee
)

DELETE
FROM DuplicateRecords
WHERE RN > 1;
```

Results | Messages

	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	102	Rahul	65000.00	1	2025-12-15
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	105	Sneha	60000.00	4	2026-03-05
6	106	Vikas	95000.00	2	2025-10-01
7	107	Neha	70000.00	5	2026-04-12
8	108	Kiran	55000.00	1	2026-05-01
9	109	Rohan	80000.00	3	2026-01-15
10	110	Pooja	68000.00	4	2026-06-01

```
—-----------------------------------------------------
-- Insert additional employees

INSERT INTO Employee
VALUES
(113,'Anjali',72000,5,'2025-03-15'),
(114,'Suresh',88000,3,'2025-02-20'),
(115,'Ravi',62000,7,'2025-01-10'),
(116,'Komal',78000,8,'2025-04-05'),
(117,'Nikhil',69000,10,'2025-05-12');
```

Results Messages

	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	102	Rahul	65000.00	1	2025-12-15
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	105	Sneha	60000.00	4	2026-03-05
6	106	Vikas	95000.00	2	2025-10-01
7	107	Neha	70000.00	5	2026-04-12
8	108	Kiran	55000.00	1	2026-05-01
9	109	Rohan	80000.00	3	2026-01-15
10	110	Pooja	68000.00	4	2026-06-01
11	113	Anjali	72000.00	5	2025-03-15
12	114	Suresh	88000.00	3	2025-02-20
13	115	Ravi	62000.00	7	2025-01-10
14	116	Komal	78000.00	8	2025-04-05
15	117	Nikhil	69000.00	10	2025-05-12

--- Window Functions

--- 1.ROW_NUMBER()

```
SELECT
    EmpID,
    EmpName,
    Salary,
    ROW_NUMBER() OVER(ORDER BY Salary DESC) AS RowNum
FROM Employee;
```

	EmpID	EmpName	Salary	RowNum
1	106	Vikas	95000.00	1
2	103	Amit	90000.00	2
3	114	Suresh	88000.00	3
4	104	Priya	85000.00	4
5	109	Rohan	80000.00	5
6	116	Komal	78000.00	6
7	101	Megha	75000.00	7
8	113	Anjali	72000.00	8
9	107	Neha	70000.00	9
10	117	Nikhil	69000.00	10
11	110	Pooja	68000.00	11
12	102	Rahul	65000.00	12
13	115	Ravi	62000.00	13
14	105	Sneha	60000.00	14
15	108	Kiran	55000.00	15

--- 2.RANK()

```
SELECT
   EmpID,
   EmpName,
   Salary,
   RANK() OVER(ORDER BY Salary DESC) AS RankNo
FROM Employee;
```

	EmpID	EmpName	Salary	RankNo
1	106	Vikas	95000.00	1
2	103	Amit	90000.00	2
3	114	Suresh	88000.00	3
4	104	Priya	85000.00	4
5	109	Rohan	80000.00	5
6	116	Komal	78000.00	6
7	101	Megha	75000.00	7
8	113	Anjali	72000.00	8
9	107	Neha	70000.00	9
10	117	Nikhil	69000.00	10
11	110	Pooja	68000.00	11
12	102	Rahul	65000.00	12
13	115	Ravi	62000.00	13
14	105	Sneha	60000.00	14
15	108	Kiran	55000.00	15

--- 3.DENSE_RANK()

```
SELECT
   EmpID,
   EmpName,
   Salary,
   DENSE_RANK() OVER(ORDER BY Salary DESC) AS DenseRankNo
FROM Employee;
```

Results | Messages

	EmpID	EmpName	Salary	DenseRankNo
1	106	Vikas	95000.00	1
2	103	Amit	90000.00	2
3	114	Suresh	88000.00	3
4	104	Priya	85000.00	4
5	109	Rohan	80000.00	5
6	116	Komal	78000.00	6
7	101	Megha	75000.00	7
8	113	Anjali	72000.00	8
9	107	Neha	70000.00	9
10	117	Nikhil	69000.00	10
11	110	Pooja	68000.00	11
12	102	Rahul	65000.00	12
13	115	Ravi	62000.00	13
14	105	Sneha	60000.00	14
15	108	Kiran	55000.00	15

--- Department Wise Salary Ranking

```
SELECT
    EmpName,
    Salary,
    DeptID,
    RANK() OVER
    (
        PARTITION BY DeptID
        ORDER BY Salary DESC
    ) AS DeptRank
FROM Employee;
```

	EmpName	Salary	DeptID	DeptRank
1	Rahul	65000.00	1	1
2	Kiran	55000.00	1	2
3	Vikas	95000.00	2	1
4	Amit	90000.00	2	2
5	Megha	75000.00	2	3
6	Suresh	88000.00	3	1
7	Priya	85000.00	3	2
8	Rohan	80000.00	3	3
9	Pooja	68000.00	4	1
10	Sneha	60000.00	4	2
11	Anjali	72000.00	5	1
12	Neha	70000.00	5	2
13	Ravi	62000.00	7	1
14	Komal	78000.00	8	1
15	Nikhil	69000.00	10	1

```
----- Find second highest salary
```

```
SELECT MAX(Salary) AS SecondHighestSalary
FROM Employee
WHERE Salary <
(
    SELECT MAX(Salary)
    FROM Employee
);
```

	SecondHighestSalary
1	90000.00

```
-- Employees Whose Salary > Department Average Salary
```

```
SELECT
    EmpID,
    EmpName,
    Salary,
    DeptID
FROM Employee e
WHERE Salary >
(
    SELECT AVG(Salary)
    FROM Employee
```

```
    WHERE DeptID = e.DeptID
);
```

	EmpID	EmpName	Salary	DeptID
1	102	Rahul	65000.00	1
2	103	Amit	90000.00	2
3	106	Vikas	95000.00	2
4	114	Suresh	88000.00	3
5	104	Priya	85000.00	3
6	110	Pooja	68000.00	4
7	113	Anjali	72000.00	5

--- INNER JOIN

```
SELECT
    e.EmpID,
    e.EmpName,
    e.Salary,
    d.DeptName
FROM Employee e
INNER JOIN Department d
ON e.DeptID = d.DeptID;
```

	EmpID	EmpName	Salary	DeptName
1	101	Megha	75000.00	IT
2	102	Rahul	65000.00	HR
3	103	Amit	90000.00	IT
4	104	Priya	85000.00	Finance
5	105	Sneha	60000.00	Sales
6	106	Vikas	95000.00	IT
7	107	Neha	70000.00	Marketing
8	108	Kiran	55000.00	HR
9	109	Rohan	80000.00	Finance
10	110	Pooja	68000.00	Sales
11	113	Anjali	72000.00	Marketing
12	114	Suresh	88000.00	Finance
13	115	Ravi	62000.00	Operations
14	116	Komal	78000.00	Support
15	117	Nikhil	69000.00	Research

```
--- LEFT JOIN

SELECT
    d.DeptID,
    d.DeptName,
    e.EmpName
FROM Department d
LEFT JOIN Employee e
ON d.DeptID = e.DeptID;
```

Results | Messages

	DeptID	DeptName	EmpName
2	1	HR	Kiran
3	2	IT	Megha
4	2	IT	Amit
5	2	IT	Vikas
6	3	Finance	Priya
7	3	Finance	Rohan
8	3	Finance	Suresh
9	4	Sales	Sneha
10	4	Sales	Pooja
11	5	Marketing	Neha
12	5	Marketing	Anjali
13	6	Admin	NULL
14	7	Operations	Ravi
15	8	Support	Komal
16	9	Legal	NULL
17	10	Research	Nikhil

```
--- GROUP BY with HAVING

SELECT
    DeptID,
    COUNT(*) AS EmployeeCount
FROM Employee
GROUP BY DeptID
HAVING COUNT(*) > 1;
```

	DeptID	EmployeeCount
1	1	2
2	2	3
3	3	3
4	4	2
5	5	2

--- Employees Hired in Last 6 Months

```
SELECT *
FROM Employee
WHERE HireDate >= DATEADD(MONTH,-6,GETDATE());
```

	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	103	Amit	90000.00	2	2026-01-20
3	105	Sneha	60000.00	4	2026-03-05
4	107	Neha	70000.00	5	2026-04-12
5	108	Kiran	55000.00	1	2026-05-01
6	109	Rohan	80000.00	3	2026-01-15
7	110	Pooja	68000.00	4	2026-06-01